

Temperature-Dependent Gene Expression

Introduction

Genes are DNA sequences that occupy specific locations on chromosomes and control the heritable traits of all living things. They encode the information to make proteins and to determine when and how much of those proteins will be made. When a gene is expressed, it is turned on and is actively producing proteins. When a gene is not expressed, it is turned off and is not actively producing protein, but is poised to begin production at any time.

We live in a constantly changing environment. Each day we are faced with many different conditions to which we must respond and adapt. Many of these responses are dictated by our genes and the way in which they are regulated. All of the cells in our body contain the same collection of many thousands of genes. However, these genes are not all active at once. Certain genes are expressed at certain times in response to changes in the internal and external environment. The ability to exhibit such regulated control of gene expression is crucial not only to people, but to all living organisms. Bacteria are small, unicellular organisms that are amazingly adaptive despite their microscopic size. They have the ability to turn the expression of certain genes on or off in response to their surrounding conditions.

Although bacteria are microscopic, single-celled organisms, they grow and divide rapidly. Each time a bacterial cell divides, it creates an exact copy—a genetic duplicate—of itself. As bacteria multiply, the resulting mass of cells becomes visible. On a solid medium, it looks like a layer of slime. Each individual cell in this mass is a discrete organism, but all of the cells are genetically identical and will respond similarly to environmental conditions.

In this activity, you will manipulate the temperature growth condition of the bacterium *Serratia marcescens* and observe how this change affects production of a cell surface pigment called prodigiosin. Prodigiosin, a red pigment, causes the bacteria to appear red when it is produced and white when it is not. This visual result will enable you to infer how the genes that control prodigiosin production are regulated at different temperatures.

Instructions

Day 1

1. Obtain two pre-poured nutrient agar plates for your group. Orient the plates so that they are lid-side-down on your table. Using the marker provided, draw a line down the center of the bottom of each plate. Label one plate "1" and the other plate "2." On plate 1, label the left side "A" and "RT" (for room temperature), and label the right side "B" and "37°C." On plate 2, label the left side "C" and "37°C," and label the right side "D" and "RT." On both plates, write your group name and the current date (see Figure 1).

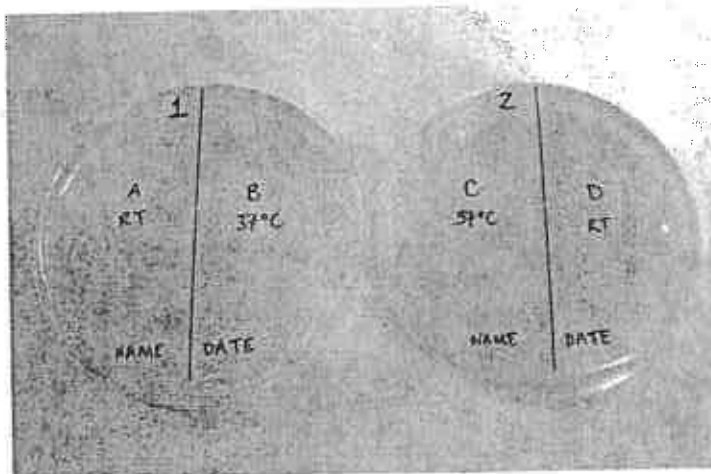


Figure 1. Plate labeling

Your teacher will provide you with a culture of *Serratia marcescens*. Have one person in your group transfer a very small amount of the culture to side A of plate 1, following the directions in steps 2–5.

2. Open the pack of sterile applicator sticks according to the directions on the package. Remove one applicator, touching only the stick. Do not touch the cotton tip or allow it to touch a non-sterile surface.
3. Lift one side of the lid of the *S. marcescens* culture plate just enough to insert the applicator stick (see Figure 2). Gently touch the cotton tip to the bacteria. You should see a small red spot on the cotton tip. Remove the applicator stick and replace the lid on the *S. marcescens* culture. Handle the applicator stick carefully and do not allow it to contact any surface.
4. Turn over plate 1 so that it is lid-side-up, and then locate side A of the plate. Lift the lid just enough so that you can access side A. Inoculate the “A” half of the plate by lightly moving the cotton tip of the applicator stick across the surface of the agar in a zigzag motion (see Figure 3). Replace the lid on the plate. (This method of inoculation is called streaking. Bacterial cells from the stock culture are deposited on the plate in small quantities. They will grow in the pattern in which they were streaked and will become visible overnight.)
5. Place the used applicator stick on the paper towel at your station. Do not disturb the applicator stick.
6. Have a different member of your group inoculate side C of plate 2, following the directions in steps 2–5.
7. Turn plates 1 and 2 lid-side-down. Leave plate 1 at your station, at room temperature, overnight. Put plate 2 in the 37°C incubator overnight.
8. Dispose of the used applicator sticks in the autoclave disposal bag provided, or according to your teacher’s instructions. Wash your hands thoroughly with soap and water.



Figure 2. Obtaining *Serratia marcescens* from the stock culture

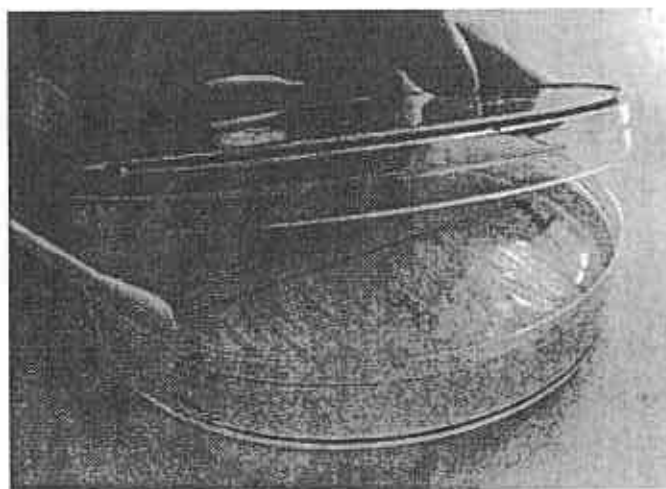


Figure 3. Bacterial streak pattern

Day 2

Today, after observing the results of the overnight incubation and making some predictions, you will inoculate side B of plate 1 and side D of plate 2, and then incubate the plates at 37°C and at room temperature, respectively (reversing the previous temperature conditions).

9. a. Obtain plates 1 and 2. Observe the bacteria by looking through the lid and through the bottom of each plate. *Do not remove the lids.* Hold each plate up to the light while you observe it. In the Observation Table on page S-4, record the Day 2 color of the bacteria on side A (RT) and side C (37°C).
- b. On the basis of your Day 2 results, predict the color of the bacteria in positions B and D on Day 3. What color do you think the bacteria in positions A and C will be on Day 3, after switching the growth temperatures? Record your predictions in the Observation Table on page S-4.

Have one person in your group transfer a very small amount of the culture to side B of plate 1, following the directions in steps 10–12.

10. Open the pack of sterile applicator sticks according to the directions on the package. Remove one applicator, touching only the stick. Do not touch the cotton tip or allow it to touch a non-sterile surface.
11. Turn over plate 1 so that it is lid-side-up. Lift the lid just enough so that you can access side B. Touch the cotton tip to the top edge of the bacterial streak in side A. Without lifting the cotton tip, drag it to side B and gently streak across the agar surface in a zigzag motion (see Figure 4). Replace the lid on the plate.
12. Place the used applicator stick on the paper towel at your station. Do not disturb the applicator stick.
13. Have a different member of your group inoculate side D of plate 2, following the directions in steps 10–12.
14. Turn plates 1 and 2 lid-side-down. Put plate 1 in the 37°C incubator overnight. Leave plate 2 at your station, at room temperature, overnight.
15. Dispose of the used applicator sticks in the autoclave disposal bag provided, or according to your teacher's instructions. Wash your hands thoroughly with soap and water.

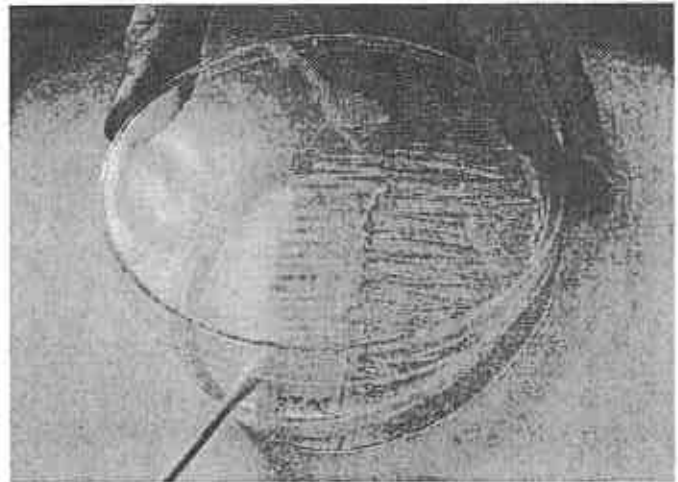


Figure 4. Bacterial second streak pattern

Day 3

16. Obtain plates 1 and 2. Observe the bacteria by looking through the lid and through the bottom of each plate. *Do not remove the lids.* Hold each plate up to the light while you observe it. Record your Day 3 observations of the color of the bacteria in position B and D in the Observation Table on page S-4. Then, record your observations of the color of the bacteria in positions A and C (after a second day of growth at a different temperature).

Observation Table

Day	Bacteria	Growth Temperature	Predicted Color	Observed Color
2	A	RT		
2	C	37°C		
3	B	37°C		
3	D	RT		
3	A	RT, then 37°C		
3	C	37°C, then RT		

Questions

1. At what growth temperature did your *S. marcescens* culture produce prodigiosin? What does this imply about the expression of prodigiosin-producing genes at this temperature?
2. What happens to the expression of prodigiosin-producing genes at 37°C? How is this apparent in the appearance of the *S. marcescens* bacteria?
3. What happened to the color of the *S. marcescens* bacteria in position A after being incubated at room temperature for one day and then at 37°C for one day?
4. What is aseptic technique and why it is used when working in a microbiology lab?